



Global Food Waste & Supply Chain Analysis

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1. Title Slide

Project: Global Food Waste & Supply Chain Analysis

Tools: Python, Pandas, Seaborn, Matplotlib, SQLite, Power BI

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Global Food Supply Chain



A large, abstract teal graphic on the left side of the slide, featuring curved, overlapping shapes that create a sense of depth and movement.

2. Problem Statement

1/3 of global food production is wasted yearly.

Leads to economic loss and environmental damage.

Goal: Identify patterns, key causes, and opportunities for reduction.



3. Objectives

1

Analyze waste by country, category, and year

2

Find correlation with population and economic economic loss

3

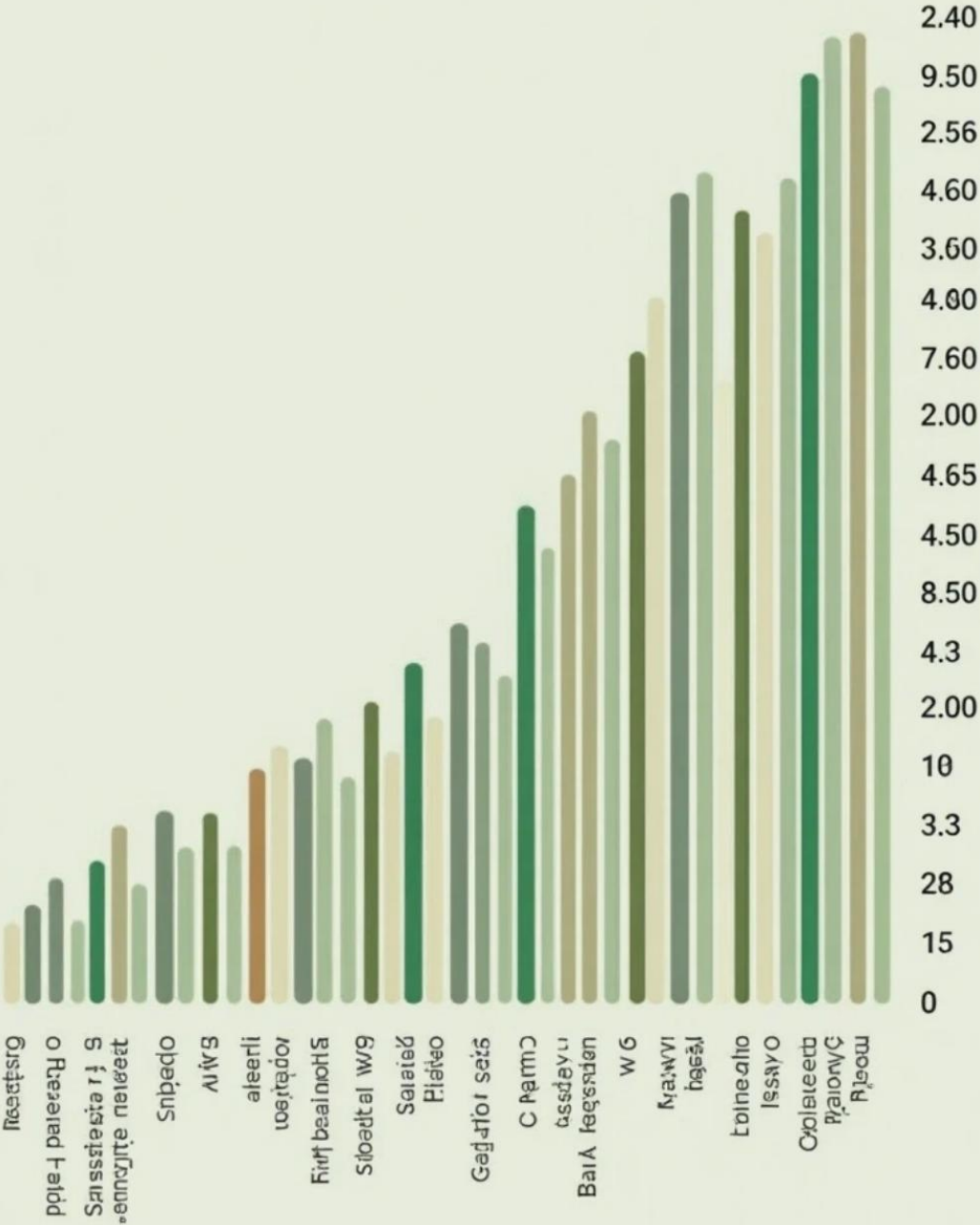
Build Power BI dashboard for sustainability insights

4. Dataset Overview

Source: FAO / Kaggle Global Food Waste Dataset

5,000 rows, 8 columns

Key features: country, year, food_category, total_waste_tons, economic_loss_million_usd, population_million



5. Data Cleaning

1

Standardized column names

2

Removed spaces & inconsistencies

3

Added derived column: waste_per_person_kg\

Formula: $(\text{total_waste_tons} * 1000) / (\text{population_million} * 1e6)$





6. Exploratory Data Analysis

Top 10 waste-producing countries

Most wasted food categories

Yearly waste trend (2000–2020)

Correlation: higher population → higher waste

7. SQL Insights

1

Used sqlite3 in Jupyter Notebook

2

Queries: top countries, food categories, yearly trends, avg waste per person

3

Integrated Python + SQL for efficient querying





8. Power BI Dashboard

Visual Components:

Cards

Total Waste, Economic Loss, Waste per Capita

Bar Chart

Top countries by waste

Column Chart

Top food categories

Line Chart

Waste trend over years

Map

Waste by country

Filters

Year, Food Category, Country

9. Key Insights

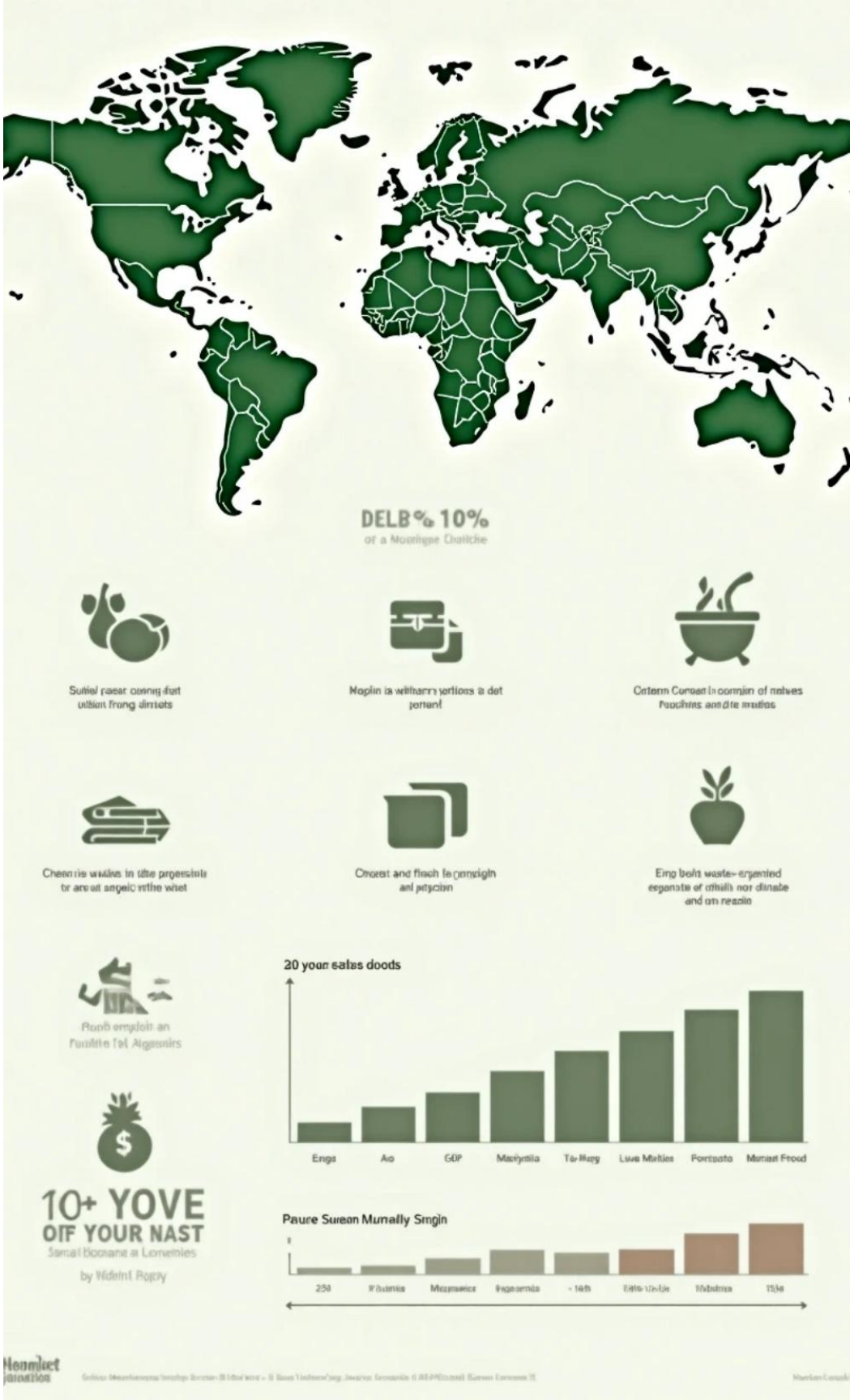
- 1

A few countries produce >50% of waste
- 2

Vegetables & Fruits → highest waste waste categories
- 3

Food waste correlates with GDP & urban population
- 4

Strong case for improving supply chain chain efficiency





10. Outcome

- ✓ Clean dataset ready for visualization
- ✓ Combined Python, SQL & Power BI
- ✓ Actionable insights for sustainability policy

11. Future Scope

1

Predictive model for future waste

2

Include CO₂ impact

3

Integrate live APIs for real-time data

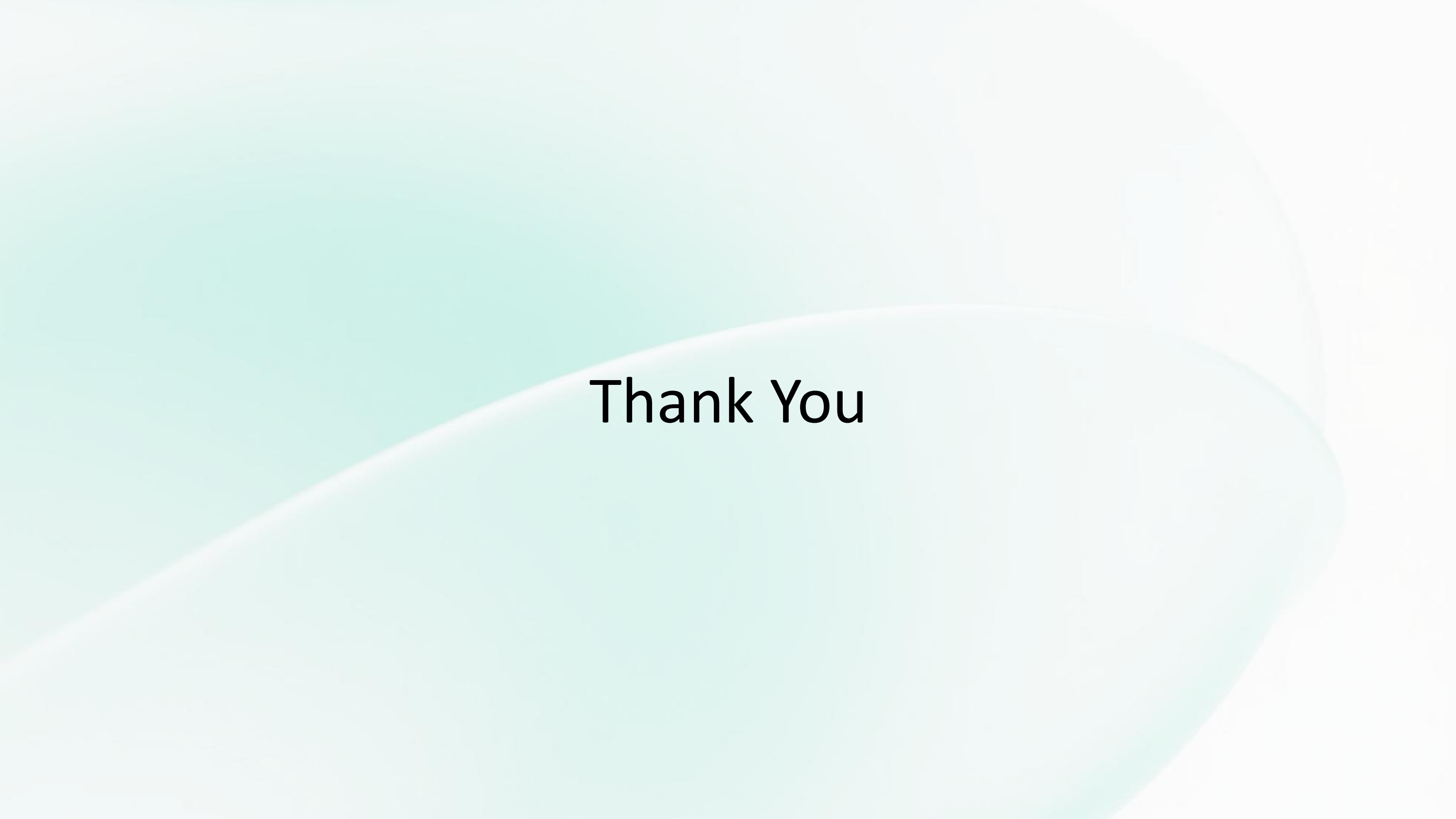


12. Thank You

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Thank You